THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 9: **Some Applications of Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is 20 m above the level of the person's eyes and the eyes are 20 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) 15° **(b)** 30° **(c)** 45° **(d)** 60°

2. From a point on level ground, 6 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) 6 m **(b)** $6\sqrt{3}$ m **(c)** 2 m **(d)** $2\sqrt{3}$ m

3. A ladder 30 m long leans against a vertical wall and makes an angle of 60° with the level ground. The height reached on the wall is: 1

(a) $15\sqrt{3}$ m **(b)** $\frac{15\sqrt{3}}{2}$ m **(c)** 15 m **(d)** $30\sqrt{3}$ m

4. A boy whose eyes are 1 m above the ground stands 20 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) 22 m **(b)** 20 m **(c)** 21 m **(d)** $21\sqrt{3}$ m

5. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 60 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) 30 m **(b)** $60\sqrt{3}$ m **(c)** $30\sqrt{3}$ m **(d)** $20\sqrt{3}$ m

6. A storm breaks a tree at a height of 9 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. The original height of the tree is: 1

(a) 27 m **(b)** $27\sqrt{3}$ m **(c)** 18 m **(d)** 54 m

7. From the top of a building 12 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. The height of the tower is: 1

(a) $12\sqrt{3}$ m **(b)** $12 + 12\sqrt{3}$ m **(c)** 24 m **(d)** $24 + 24\sqrt{3}$ m

8. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is 25 m above the level of the person's eyes and the eyes are 25 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) 15° **(b)** 45° **(c)** 60° **(d)** 30°

9. From a point on level ground, 35 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $\frac{35\sqrt{3}}{3}$ m **(b)** $35\sqrt{3}$ m **(c)** $70\sqrt{3}$ m **(d)** 35 m

10. A ladder 18 m long leans against a vertical wall and makes an angle of 60° with the level ground. The height reached on the wall is: 1

(a) $18\sqrt{3}$ m **(b)** $9\sqrt{3}$ m **(c)** 9 m **(d)** $\frac{9\sqrt{3}}{2}$ m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is $14\sqrt{3}$ m above the level of the person's eyes and the eyes are 14 m horizontally from the pole. Find the angle of elevation of the top of the pole. **2**

12. A storm breaks a tree at a height of 15 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. Find the distance from the foot of the tree to the point where its top touches the ground. **2**

13. A ladder 36 m long leans against a vertical wall and makes an angle of 60° with the level ground. Find the distance of the foot of the ladder from the wall. **2**

14. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is $16\sqrt{3}$ m above the level of the person's eyes and the eyes are 16 m horizontally from the pole. Find the angle of elevation of the top of the pole. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The shadow of a vertical tower on level ground is 24 m longer when the sun's elevation is 30° than when it is 60°. Find the height of the tower. **3**

16. From the top of a building 16 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower. **3**

17. From the top of a tower 100 m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 45°. Find the distance between the two objects. **3**

OR

The shadow of a vertical tower on level ground is 60 m longer when the sun's elevation is 30° than when it is 60°. Find the height of the tower. **3**

18. From the top of a building 20 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower. **3**

OR

From the top of a tower 30 m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 45°. Find the distance between the two objects. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two vertical poles of equal height stand on opposite sides of a road 200 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). **5**

OR

A statue 10 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. Find the height of the pedestal. **5**

20. Two vertical poles of equal height stand on opposite sides of a road 240 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). **5**

OR

A statue 12 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. Find the height of the pedestal. **5**

END OF QUESTION PAPER — Total: 40 Marks